Yash Dave

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Education

Stanford University 2025–2027 (expected)

M.S., Computational and Mathematical Engineering, Data Science Track

GPA: 3.9

Coursework: Numerical Linear Algebra, Applied Statistics, ML from Human Preferences, Convex Optimization, ML Benchmarks & Competitions

University of California, Berkeley 2021–2025

B.A., Applied Mathematics (Economics cluster) and Computer Science (Honors); Public Policy Minor

GPA: 3.94

Coursework: ML*, Optimization*, Computer Vision*, Algorithms, Linear Algebra, Probability, Statistical Inference *graduate-level

Publications & Works in Progress

Y. Dave*, S. Truong*, S. Wang†, S. Koyejo†. *The AI Evaluation Ecosystem* . arXiv:2610.09296. AI Measurement Science Workshop, COLM 2026. *equal contribution, †equal advising

M. Hardy, A. Reuel, L. Zhang, J. M. Casabianca, S. Truong, **Y. Dave**, H. Lee, B. W. Domingue, S. Koyejo. *AI Cartography: Mapping the Latent Landscape of AI Benchmark Ecosystems* . ICML 2026.

B. Marks*, **Y. Dave***, Z. Wang, H. Chung, R. Patwa, S. Cha, M. Murphy, A. Strang. *Do Generalized-Gamma Scale Mixtures of Normals Fit Large Image Datasets?* . arXiv:2512.17038, under review. *equal contribution

Research Experience

Snowflake **Menlo Park, CA**

Research Intern June–August 2026

- Designed and shipped a user-in-the-loop method that mines sessions to jointly improve memory, skills, hooks, and MCP
- Built multi-turn memory evals and harness-agnostic benchmark infra for Claude Code, Codex, OpenCode, and OpenHands
- Turned real agent sessions into an internal database-engineering benchmark graded by a calibrated LLM judge

Stanford Trustworthy AI Research Lab **Stanford, CA**

Research Assistant (under Prof. Sanmi Koyejo) June 2025–present

- Building an expert-validated generative agent-based model of the AI eval ecosystem to study evaluator and policy design
- Exploring methods for sample-efficient prediction of agent reliability (e.g., cold-start IRT, TabPFN)

Strang Lab **Berkeley, CA**

Research Assistant (under Prof. Alexander Strang) January 2024–September 2025

- Developed a testing pipeline to evaluate realism of a Bayesian mixture model for modeling quasi-sparse image data
- Created a performant library for generating CDFs of prior distributions at scale, finding best-fit parameters, and visualization

UC Berkeley School of Information **Berkeley, CA**

Research Assistant (under Profs. Niloufar Salehi and Afshin Nikzad) September 2023–December 2023

- Built and evaluated retrieval strategies for a personal information management system, combining semantic and keyword search with Ragas, improving suggestion faithfulness and relevance by 15% and 10% respectively
- Designed and shipped a SearchableDropdown question-answering interface that increased user engagement by 20%

Meet Your Future Project (Berkeley Economics and BRAC Uganda) **Berkeley, CA**

Research Apprentice (under PhD Candidate Livia Alfonsi and Prof. Jeremy Magruder) January 2022–October 2022

- Achieved 83% acc. in sentence-level topic classification of a 500+ transcript dataset using NLI-based zero-shot classification
- Performed sentiment analysis and emotion detection using VADER and NRC dictionary, creating visualizations using ggplot2

Talks & Posters

AI Measurement Science Workshop (*Toward Rigorous AI Evaluation*), COLM 2026, San Francisco. *The AI Evaluation Ecosystem* (poster), October 9, 2026. [workshop]

CS Research Lightning Talks, Stanford. *The AI Evaluation Ecosystem*, September 29, 2026.

ICME Research Symposium, Stanford. *The AI Evaluation Ecosystem*, May 19, 2026. [poster]

AI+Science: Accelerating Discovery Conference, Stanford HAI. *The AI Evaluation Ecosystem*, May 5, 2026. [poster] [event]

Stanford Trustworthy AI Research (STAIR) Lab, AI Measurement Science Reading Group. *The AI Evaluation Ecosystem*, April 3, 2026.

Joint Statistical Meeting, Section on Bayesian Statistical Science. *Do Sparsity-Promoting Hierarchical Prior Models Plausibly Model Empirical Image Data*, August 5, 2025. [poster] [event]

UC Berkeley, South Asian Politics (PolSci 145B). *Beliefs about Government Intervention in India*, May 12, 2025. [poster]

MPS Scholars Retreat (3-min SLAM), UC Berkeley. *Do Sparsity-Promoting Hierarchical Prior Models Plausibly Model Empirical Image Data*, April 4, 2025.

Teaching

Course Assistant, CS 229 Machine Learning , Stanford Winter 2026

- Mentor student projects on ML modeling and evaluation; grade assignments and exams; hold office hours

Course Assistant, CS 283 Governing Artificial Intelligence , Stanford Autumn 2025

- Developed rubrics and graded technical AI evaluation assignments, policy memos, and philosophy papers

UC Berkeley College of Computing, Data Science and Society **Berkeley, CA**

Data 140 (Probability for Data Science) and Data 375 (Pedagogy) Teaching Assistant August 2024–May 2025

- Hold two weekly sections; handle Data 375 logistics, work with professors to design session outlines, and facilitate discussion

Data 100 (Data Science) Lead Content TA Jan 2023–May 2024

- Developed and improved content, oversaw beta-testing and directed release of assignments for 1200+ students on JupyterHub
- Incorporated human contexts and ethics (HCE) material with regression/classification projects and discussion worksheets
- Taught 1-2 discussion sections to 20-45 students weekly for 4 semesters; rated 6.55/7 overall effectiveness (n=81)
- Co-authored comprehensive debugging guide and data science and ML reference text, reducing office hour tickets by 40%

Honors & Awards

Phi Beta Kappa, UC Berkeley 2025
High Honors in General Scholarship, UC Berkeley 2025
Outstanding GSI Award, UC Berkeley Computing, Data Science and Society 2025
Upsilon Pi Epsilon, Computer Science Honor Society 2025
Dean's List, UC Berkeley 2022–2025
Kevin K. Gong Memorial Scholarship, UC Berkeley EECS 2024
EECS Honors Program, UC Berkeley 2025
World Bank Group Youth Summit Case Challenge, Honorable Mention 2023
Citadel West Coast Regional Datathon, Honorable Mention 2023
EA Berkeley AI Safety Distillation Contest, 3rd Place 2022

Selected Projects

Direct Preference Optimization applied to Financial Advising December 2025

- Fine-tuned a 4B-parameter model using DPO to automate SEC Marketing Rule compliance review, reducing reasoning tokens by 50% while maintaining expert-level rewrite quality
- Built a privacy-preserving synthetic data pipeline generating 943 preference pairs from 40 expert annotations using a teacher-fixer-breaker framework to train compliant ad reviewers

Service, Leadership & Outreach

President, Tech+Social Impact Club May 2023–May 2024

- Co-facilitate student-run course about data ethics, technology and social impact to a total of 100 students over 3 semesters
- Organized a tech help session at Next Village SF, helping 20-30 elders and saving the nonprofit an estimated $1200

Advising Fellow, Matriculate Jan 2022–Aug 2023

- Mentored three first-gen high school juniors throughout the college application process: reviewing essays, scholarships etc.

Data Science and Philosophy through the Spider-verse — Splash @ Stanford (Nov 16, 2025; Apr 11, 2026); Splash @ Berkeley (Apr 13, 2025).

Technical Skills

Python (NumPy, pandas, PyTorch, JAX), R, SQL, C/C++, Go, Java; Git, LaTeX, AWS, CUDA

Languages

English (professional working), Hindi (limited working), Spanish (limited working), Arabic (elementary)